

Audio-Agent: Bridging the Semantic Gap in Neural Audio Effects via Multi-Modal Retrieval-Augmented Generation

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Abstract—Digital audio workstations (DAWs) expose rich effect chains, yet a semantic gap remains between perceptual user intent (e.g., “warm” or “spacious”) and low-level DSP control parameters. Waveform-generative models can synthesize realistic audio, but the resulting samples are difficult to edit within DAW-centric workflows. We present Audio-Agent, a neuro-symbolic framework that generates editable plugin parameter configurations instead of finalized waveforms.

Audio-Agent combines three components: (1) Texture Resonance Retrieval (TRR), which represents audio texture with second-order feature statistics (Gram matrices) computed from frame-level embeddings; (2) uncertainty-aware dual-modal fusion, which reweights text and audio evidence based on retrieval uncertainty; and (3) constrained parameter projection, which deterministically repairs LLM drafts to satisfy plugin feasibility constraints.

In objective parameter-space evaluation on a fixed 30-query held-out split, Audio-Agent improves parameter alignment over first-order retrieval baselines and remains robust under single-modality degradation. A MUSHRA-style listening study with 26 participants further supports the practicality of retrieval-grounded parameter control for production workflows.

I. INTRODUCTION

The field of neural audio synthesis currently faces a fundamental tension between *fidelity* and *control*. Recent high-fidelity generative models, primarily based on latent diffusion [1, 2], can achieve strong perceptual quality but often operate as “black boxes.” They generate finalized waveforms where internal attributes—such as the attack time of a compressor or the decay of a reverb—are entangled and inaccessible for post-hoc editing. Conversely, Differentiable Digital Signal Processing (DDSP) [3] offers interpretable, structured control but often suffers from the ill-posed nature of the inverse parameter estimation problem.

This tension motivates the development of **Neuro-Symbolic Audio Agents**: systems that leverage the reasoning capabilities of Large Language Models (LLMs) to control interpretable DSP engines. By predicting parameters rather than samples, such systems enable fully editable outputs compatible with professional production workflows.

However, realizing this paradigm requires overcoming two critical challenges. First, LLMs trained on discrete tokens struggle to regress continuous values within strict physical constraints, leading to a *hallucination problem* where generated parameters violate DSP valid ranges (e.g., negative filter frequencies) or inter-parameter dependencies. Second, grounding the LLM via retrieval requires a similarity metric that captures *timbral texture*. Standard audio representations (e.g., CLAP-style embeddings or mean-pooled Wav2Vec2 features) summarize an audio segment into a single vector; while effective for coarse semantics, such pooling can discard

temporal co-activation patterns that are critical for texture-dominant effects (e.g., tremolo modulation or signal-dependent distortion).

To bridge these gaps, we present **Audio-Agent**, a framework that frames parameter generation as a *retrieval-grounded reasoning task*. Our system retrieves validated presets based on user intent and adapts them via an LLM to satisfy user queries. To address the texture representation gap, we draw inspiration from the neuroscience of sound texture perception [4] and visual style transfer [5], introducing **Texture Resonance Retrieval (TRR)**. TRR models audio style using Gram matrices of deep feature activations, explicitly capturing the *second-order correlations* that define temporal texture.

Our key contributions are: (i) **Texture-Aware Grounding**: we observe that second-order statistics (Gram matrices) can improve retrieval for texture-dominant effects, yielding better parameter alignment on our benchmark; (ii) **Uncertainty-Aware Modal Fusion**: we propose an entropy-based fusion mechanism that adjusts the relative weight of text and audio retrieval under degraded inputs; and (iii) **Constrained Neuro-Symbolic Execution**: we introduce a deterministic constraint projection layer that enforces plugin-defined validity constraints on LLM-generated parameters.

II. RELATED WORK

Our work lies at the intersection of intelligent music production, neural audio synthesis, and LLM-based agent systems. We review each area and position our contributions accordingly.

Intelligent Music Production.

The challenge of translating perceptual descriptions into technical parameters was formally characterized by Stables et al. [6]. Their SAFE project demonstrated the feasibility of semantic-to-parameter mapping in a DAW context, but did not resolve the generalization problem across instruments, effect chains, and stylistic sub-genres. Schedl et al. [7] provides a comprehensive survey of music information retrieval techniques, highlighting semantic auto-tagging as a key challenge.

Early work in interactive control focused on learning intuitive mappings for synthesizers. Kulić et al. [8] proposed active learning with Gaussian processes to discover perceptual control knobs, while more recently, Bryan et al. [9] introduced SynthScribe, a multimodal tool for synthesizer sound retrieval. While these works advance interface design, they primarily target synthesis parameters (e.g., oscillator shape) rather than the complex signal chains of audio effects.

Audio Analysis Approaches. Subsequent work pursued automation via audio content analysis and differentiable mixing

systems [10–12]. While effective when a target mix or reference audio already exists, this direction implicitly assumes the target audio is given. It therefore struggles to support creative timbre design from a purely conceptual request (“make it more vintage breakup”) and often suffers from an **opacity problem**: users see outputs but lack interpretable, editable rationale.

Recent advances in differentiable DSP have enabled neural approaches to audio effect modeling. Peladeau et al. [13] proposes blind estimation of audio effect parameters using auto-encoder architectures, addressing the inverse problem of parameter recovery. Liu et al. [14] introduces DDSP-SFX, which leverages differentiable digital signal processing for acoustically-guided sound effects generation, combining neural networks with traditional DSP modules for controllable synthesis. De Man et al. [15] explores generative latent spaces for neural audio synthesis, bridging DSP-informed and data-driven approaches. In parallel, style transfer methods have been applied to audio effects, with Benetos et al. [16] introducing ST-ITO for controlling audio effects via inference-time optimization using pretrained audio representations.

The LLM Frontier. Recent work explores LLMs as natural language interfaces for effect parameters [17–19]. While demonstrating surprising capability, these approaches remain vulnerable to hallucination, potentially producing invalid or incoherent parameters in zero-shot settings [17]. Dialogue-based disambiguation can reduce ambiguity but shifts the burden to the user and does not scale to complex multi-effect chains.

Our Positioning. Unlike prior work that treats parameter prediction as a *regression problem* or a purely generative problem, we frame it as a *retrieval-grounded reasoning problem*: retrieval supplies concrete, previously validated exemplars, and generation becomes an adaptation step rather than invention.

Neural Audio Synthesis.

Neural audio synthesis has advanced rapidly, enabling high-fidelity text-to-audio generation [1, 2, 20, 21]. Diffusion models have emerged as the dominant paradigm, with Huang et al. [22] introducing prompt-enhanced text-to-audio generation, Wang et al. [23] achieving efficient high-quality synthesis via latent consistency models, and Liang et al. [24] exploring instruction-guided latent diffusion for better controllability. However, these models primarily output *finalized waveforms*, which are difficult to edit within professional workflows.

Accordingly, we view parameter-level generation as a **complement to, not a replacement for**, waveform-level synthesis: it targets controllable editing and iterative refinement inside DAWs.

This motivates what we call **Parameter–Waveform Duality**: waveform-level generation offers maximal expressivity but limited editability, whereas parameter-level control preserves transparency and iterative refinement at the cost of requiring domain knowledge.

In parallel, differentiable DSP (DDSP) [3] demonstrates that learned models can control interpretable synthesis parameters, and self-supervised audio representation learning (e.g., Wav2Vec 2.0 [25]) provides strong acoustic embeddings. Audio-Agent leverages these representations for similarity

TABLE I: Comparison of waveform-level and parameter-level approaches in neural audio workflows. Each paradigm offers complementary strengths for different production contexts.

Level	Primary Strength	Primary Limitation
Waveform-level	High-fidelity synthesis	Hard to edit in DAWs
Parameter-level	Fully editable control	Requires expert priors

grounding, but targets editable parameter outputs compatible with DAW-centric workflows.

Audio Representation Learning. Recent advances in audio representation learning have produced strong embeddings for semantic tasks. CLAP [26] achieves robust text-audio alignment through contrastive pretraining, while PaSST [27] and PANNs [28] provide powerful acoustic features for classification and tagging. While these representations excel at semantic understanding, they are primarily designed for classification and captioning tasks. Our work adapts audio representations (Wav2Vec2) for a different purpose: capturing timbral texture for parameter retrieval, where second-order statistics (Gram matrices) complement the first-order semantic embeddings. We use mid-level Wav2Vec2 features rather than CLAP-style single-vector embeddings because frame-level activations enable Gram matrix computation for texture characterization.

LLM Agents and Retrieval-Augmented Generation.

Retrieval-augmented generation (RAG) [29] improves reliability by grounding generation in external memory. Multimodal variants extend this idea beyond text [30, 31], with recent developments including MMKB-RAG [32] for multimodal knowledge-based systems, and frameworks for visually-rich document understanding [33]. Instruction manual understanding approaches [34] demonstrate the value of cross-modal grounding in practical applications. Agentic tool-use frameworks (e.g., [35–37]) enable language models to act through structured interfaces.

Although our focus is parameter generation, we acknowledge the rapid progress in neural audio synthesis. Recent models like AudioLDM 2 [20], AudioCraft [38], and industrial systems like Suno illustrate the power of end-to-end generation. However, these systems generally do not expose the *internal signal chain* for fine-grained editing, which remains the domain of parameter-based approaches.

a) Perceptual similarity vs. parameter transferability.:

Even strong general-purpose audio representations (e.g., CLAP- or PaSST-style embeddings) face a fundamental challenge: perceptual similarity does not necessarily imply that the underlying effect parameters are transferable. While recent audio-language models have improved zero-shot classification and captioning [26, 39–41], they primarily target discrete prediction tasks rather than continuous parameter estimation. Recent instruction-following audio language models further broaden audio-text understanding [42, 43], but executable effect-chain control remains distinct: they do not directly expose plugin constraints or guarantee feasibility of continuous control parameters. Our work addresses this gap by using second-order statistics (Gram matrices) to capture texture-relevant features that better align with parameter-space prox-

imity.

Audio Texture and Semantic Audio Retrieval.

The concept of audio texture has been studied extensively in the context of style transfer and synthesis. Gatys et al. [5] introduced Gram matrix-based style representations for images, which were later adapted to audio by Stylianou et al. [44] and others. Recent surveys on audio texture analysis [45] and audio representations for sound synthesis [46] highlight the importance of capturing second-order statistical features for timbre characterization. In contrast to prior uses primarily for texture synthesis or style transfer, we use Gram statistics as a retrieval embedding to locate executable parameter exemplars, treating retrieval as a prior for downstream constrained reasoning.

In semantic audio retrieval, the WikiMuTe dataset [47] provides a rich resource for connecting natural language descriptions with music audio, while retrieval-guided approaches [48] demonstrate the value of combining retrieval with generation for music captioning. Retrieval-augmented text-to-music generation [49] further validates the effectiveness of RAG approaches in audio tasks.

Our Texture Resonance Retrieval (TRR) builds upon these insights by using Gram matrices to capture audio texture through second-order statistics, enabling more style-consistent retrieval for parameter inference.

Table II summarizes our positioning. Our distinctive contribution is combining (1) *dual-modal* grounding for parameter generation, (2) *texture-aware* audio retrieval via TRR, and (3) *uncertainty-aware* fusion for robustness under degraded inputs.

III. PROBLEM FORMULATION

We formalize neuro-symbolic audio control as a constrained parameter synthesis problem: given a user query and an input signal, produce a parameter vector that is (i) *semantically consistent* with the user’s intent and (ii) *executable* by a real-time DSP engine. Unlike waveform generation, parameter synthesis must respect strict plugin-defined validity constraints, and the inverse mapping from perceptual language (e.g., “warm”, “punchy”) to a high-dimensional control vector is inherently ill-posed. Our goal is therefore not to claim a globally optimal solution in Θ , but to produce a feasible point estimate that is well-defined, interpretable, and grounded by concrete executable prototypes.

A. Signal Processing Dynamics

Consider a (potentially non-differentiable) DSP rendering operator $\Phi : \mathcal{X} \times \Theta \rightarrow \mathcal{X}$, where $\mathcal{X}$ denotes the space of discrete-time audio signals and $\Theta \subset \mathbb{R}^d$ denotes the d -dimensional control parameter space. Given an input signal $\mathbf{x}_{\text{in}} \in \mathcal{X}$, the operator produces an output $\mathbf{x}_{\text{out}} = \Phi(\mathbf{x}_{\text{in}}; \theta)$. In practical DAW and plugin systems, Φ can include nonlinearities, conditional branches (e.g., bypass logic), and internal state (e.g., smoothing), making end-to-end differentiation unreliable or inapplicable for direct optimization over θ .

The parameter space Θ is not a free vector space but a bounded hyper-rectangle subject to structural constraints:

$$\Theta = \{\theta \in \mathbb{R}^d \mid \mathbf{l} \preceq \theta \preceq \mathbf{u}, \quad \mathcal{C}(\theta) = 1\} \quad (1)$$

where $\preceq$ denotes element-wise inequality, $\mathbf{l}, \mathbf{u}$ define physical operating ranges (e.g., frequency bounds $[20, 20\text{k}]$), and $\mathcal{C} : \mathbb{R}^d \rightarrow \{0, 1\}$ represents a binary validity function encoding inter-parameter dependencies (e.g., ensuring filter resonance $Q > 0$). In practice, DSP parameter spaces can be non-convex and partially discrete (e.g., on/off switches or mode selectors), so we use Eq. 1 as a pragmatic executable abstraction rather than a claim of convexity or smoothness. This motivates our reliance on retrieval-grounded synthesis and deterministic validity enforcement instead of direct gradient-based optimization over Θ .

B. Goal and Constraints

The user’s intent is encapsulated in a multi-modal query $q = (t, \mathbf{a}_{\text{ref}})$, comprising a natural language description $t \in \mathcal{T}$ and an optional audio reference $\mathbf{a}_{\text{ref}} \in \mathcal{X} \cup \{\emptyset\}$. Our objective is to produce a *valid* parameter configuration that is consistent with the query:

$$\tilde{\theta} \in \Theta \quad \text{given } q, \quad (2)$$

where validity is defined by the plugin-specific constraint function $\mathcal{C}(\theta) = 1$ and physical bounds $\mathbf{l} \preceq \theta \preceq \mathbf{u}$ (Eq. 1). We use θ^* to denote an ideal (possibly non-unique) solution that best matches the user’s perceptual intent; in practice we only have access to imperfect retrieval and LLM reasoning, and therefore aim for a feasible point estimate $\tilde{\theta}$.

a) Operational objective (without probabilistic semantics).: The query-consistency of $\tilde{\theta}$ is not directly measurable as a differentiable loss, because the target is perceptual and depends on a complex signal chain. Accordingly, we adopt a retrieval-grounded operational objective: construct an executable neighborhood in Θ using a finite preset database, and then adapt within that neighborhood while enforcing feasibility. This yields a system whose intermediate artifacts (retrieved prototypes, draft parameters, constraint repairs) are auditable by design, which is critical for a journal setting emphasizing system rigor and reproducibility.

C. The Retrieval-Reasoning Decomposition

Because the inverse mapping from perceptual intent to parameters is ill-posed and the DSP validity constraints are strict, we implement a retrieval-grounded synthesis pipeline. Let z denote a retrieved preset (a discrete prototype) from a finite database. We first retrieve top- K candidates conditioned on q , then use the LLM to generate a draft parameter vector $\hat{\theta}$ conditioned on q and the retrieved context, and finally project the draft to the feasible set:

$$\tilde{\theta} = \Pi_{\Theta}(\hat{\theta}), \quad \hat{\theta} = \text{LLM}(q, \text{Top-}K(q)). \quad (3)$$

The projection operator Π_{Θ} enforces both range constraints and schema constraints (Section IV-D). Intuitively, retrieval supplies a coarse *mode-seeking* step that places the query in a plausible neighborhood of Θ , while the LLM performs a *local* adaptation step that refines parameters within that neighborhood under query-specific constraints.

TABLE II: Positioning Against Prior Work

Work	Input	Output	Grounding	Editability
SAFE [6]	Text	Parameters	None	Yes
DDSP [3]	Audio	Parameters	None	Yes
LLM2Fx [17]	Text	Parameters	None	Yes
AudioLDM [1]	Text	Waveform	Pretrained	No
HM-RAG [31]	Text+Doc	Text	Retrieval	N/A
Audio-Agent (Ours)	Text+Audio	Parameters	Dual-Modal RAG	Yes

a) Design Principle 1 (Well-defined synthesis under missing modalities).: Because $\mathbf{a}_{\text{ref}}$ is optional, the synthesis pipeline must remain well-defined when $\mathbf{a}_{\text{ref}} = \emptyset$. Our formulation treats q as a tuple with an explicit null element and requires every downstream stage (retrieval, fusion, and projection) to specify a deterministic fallback behavior when an input modality is absent (Section IV-C).

IV. METHODOLOGY

We implement the retrieval-grounded synthesis framework defined in Section III via a three-stage pipeline: (1) **Texture-Aware Retrieval** to retrieve candidate presets; (2) **Uncertainty-Guided Fusion** to combine multi-modal retrieval evidence; and (3) **Constrained Reasoning** to generate a draft $\hat{\theta}$ and project it into the valid set Θ .

A. Neuro-Symbolic Architecture

The system realizes the operator $\Phi(\mathbf{x}_{\text{in}}; \theta)$ through a decoupled design, visualized in Figure 1.

The architecture consists of three components. (i) **The real-time DSP engine** implements Φ as a low-latency audio processing graph and executes only validated parameters. We denote the parameter state consumed by the DSP engine as θ_{safe} , which can be smoothed over time to avoid zipper noise and audible artifacts. (ii) **The inference agent** implements the mapping $q \rightarrow \hat{\theta}$ by running retrieval and LLM adaptation asynchronously at a lower rate than the audio callback. (iii) **The validity guard** implements the constraint function $\mathcal{C}(\theta)$ (Eq. 1) as a deterministic gatekeeper: any draft $\hat{\theta}$ must be mapped to $\tilde{\theta} \in \Theta$ before it is allowed to update θ_{safe} .

a) Design Principle 2 (Runtime isolation).: At inference time, the real-time audio path is a deterministic function of $(\mathbf{x}_{\text{in}}, \theta_{\text{safe}})$ through $\Phi(\mathbf{x}_{\text{in}}; \theta_{\text{safe}})$. The inference agent does not execute inside the real-time callback and can only influence audio through asynchronous updates of θ_{safe} after validity checking.

B. Texture Resonance Retrieval (TRR)

The core challenge in texture-aware retrieval lies in computing a similarity metric for the audio modality that preserves effect-relevant dynamics. Standard embeddings (e.g., mean-pooled Wav2Vec2 features) collapse temporal structure into a static vector, losing information that is critical for effects like modulation and distortion.

To address this, we introduce **Texture Resonance Retrieval (TRR)**. Drawing on the hypothesis that audio texture is

Algorithm 1 Texture Resonance Retrieval (TRR) Encoding

Require: Audio sample a , Wav2Vec2 model $\mathcal{W}$

Ensure: Texture embedding $\mathbf{G}_{\text{norm}}$

1. Preprocessing: Normalize a and pad to minimum length.

2. Feature Extraction:

$\mathbf{H} \leftarrow \mathcal{W}_{\text{layer } \ell}(a) \in \mathbb{R}^{C \times T_{\text{frm}}}$

3. Gram Matrix Computation:

$\mathbf{G} \leftarrow \frac{1}{T_{\text{frm}}} \mathbf{H} \mathbf{H}^T \in \mathbb{R}^{C \times C}$

4. Normalization:

$\mathbf{G}_{\text{norm}} \leftarrow \mathbf{G} / \|\mathbf{G}\|_F$

// Frobenius norm

return $\mathbf{G}_{\text{norm}}$

encoded in second-order feature statistics [4], we compute the Gram matrix of deep activations.

Let $\mathbf{H} \in \mathbb{R}^{C \times T_{\text{frm}}}$ be the activation map from an intermediate layer ℓ of a pre-trained Wav2Vec2 model, where C is the channel depth and T_{frm} is the temporal dimension. In our experiments we use Wav2Vec2 Base and set $\ell = 10$ (Section V). The texture descriptor $\mathbf{G} \in \mathbb{R}^{C \times C}$ is defined as the time-averaged correlation matrix (Algorithm 1):

$$\mathbf{G} = \frac{1}{T_{\text{frm}}} \mathbf{H} \mathbf{H}^T \quad (4)$$

$\mathbf{G}_{ij}$ captures the co-activation of feature channels i and j irrespective of their absolute temporal position. This time-invariance allows TRR to match textures (e.g., “fast tremolo”) even if the reference and database entries are not phase-aligned. We normalize $\mathbf{G}$ via the Frobenius norm for scale invariance: $\mathbf{G}_{\text{norm}} = \mathbf{G} / \|\mathbf{G}\|_F$. Similarity is then computed as cosine similarity between flattened $\mathbf{G}_{\text{norm}}$ vectors. Computationally, Gram-matrix construction scales as $O(C^2 \cdot T_{\text{frm}})$, which is more expensive than mean pooling ($O(C \cdot T_{\text{frm}})$); in our setting with short audio snippets, this overhead is manageable and can be further reduced via caching and approximate indexing.

a) Properties and limitations.: TRR can be viewed as a second-order summary of deep features: it retains cross-channel correlation structure while discarding absolute temporal alignment. This is desirable for retrieval in effect spaces where perceptual “style” is often expressed through repeated modulation patterns or stationary texture rather than a single time-locked event. The Frobenius normalization further reduces sensitivity to global scale changes in feature activations, making cosine similarity comparisons less dominated by magnitude. However, TRR is not a universal representation: by aggregating over time, it intentionally discards fine-

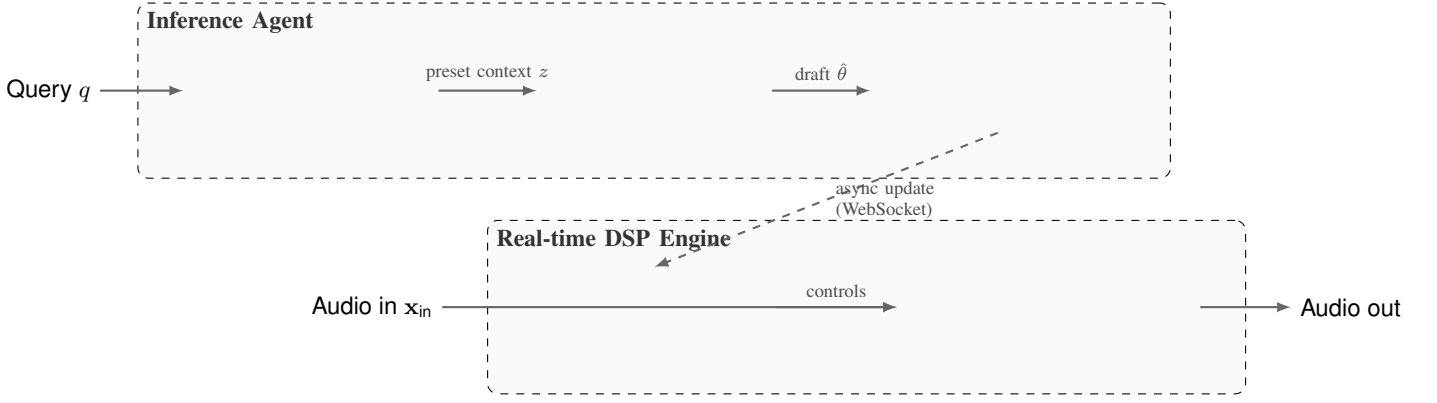


Fig. 1: The Neuro-Symbolic Architecture. The inference agent (top) retrieves candidate presets and adapts them with an LLM, while the deterministic DSP engine (bottom) executes real-time audio processing. The projection operator Π_{Θ} enforces plugin-defined validity constraints before parameters reach the real-time engine.

grained ordering and can under-represent transient-dominant effects. Moreover, as with other deep representations, retrieval behavior can depend on the chosen backbone layer and the upstream feature extractor.

b) Property 3 (Alignment-insensitive texture statistics):

Let $\mathbf{H} \in \mathbb{R}^{C \times T_{\text{fm}}}$ be the frame-level feature matrix and let $\mathbf{P} \in \mathbb{R}^{T_{\text{fm}} \times T_{\text{fm}}}$ be any permutation matrix over frames. Then the TRR Gram matrix satisfies $\frac{1}{T_{\text{fm}}} \mathbf{H} \mathbf{H}^T = \frac{1}{T_{\text{fm}}} (\mathbf{H} \mathbf{P}) (\mathbf{H} \mathbf{P})^T$. Therefore, TRR depends on time-aggregated co-activations and is insensitive to the absolute alignment of frames, which motivates its use as a texture-oriented retrieval signal.

C. Uncertainty-Aware Modal Fusion

The query q often contains asymmetric information. For example, a vague text prompt can produce a flat text-retrieval score distribution (high uncertainty), while the audio reference $\mathbf{a}_{\text{ref}}$ can still yield a peaked audio-retrieval distribution (low uncertainty). We propose a dynamic fusion strategy that weighs modalities by their *retrieval confidence*.

a) Score calibration and preference distributions: For each modality, we retrieve top- K candidates with raw similarity scores (e.g., cosine, dot-product, or BM25). Because score scales can differ across retrievers, any score-level fusion requires a calibration step; otherwise, the fused score can be dominated by an arbitrary scale rather than evidence. We apply a minimal calibration by normalizing the top- K scores into $[0, 1]$ via min-max normalization before computing the entropy of the resulting softmax preference distribution. Concretely, for a score vector $\mathbf{r}$ we compute $s_i = (r_i - r_{\min}) / (r_{\max} - r_{\min} + \epsilon)$, where ϵ avoids division-by-zero when the top- K scores are nearly constant. We then convert the calibrated scores into a *normalized preference distribution* $p^{(m)}$ via a unit-temperature softmax. Importantly, $p^{(m)}$ is not interpreted as a probabilistic posterior; it is a device for measuring how concentrated the modality’s retrieval preference is over the top- K list. The modality weight is then computed via (Algorithm 2):

$$w_m \propto \exp \left(-\beta \cdot \frac{H(p^{(m)})}{\log K} \right) \quad (5)$$

Algorithm 2 Uncertainty-Aware Modal Fusion

Require: Query $q = (t, \mathbf{a}_{\text{ref}})$, KBs $\mathcal{K}_{\text{text}}, \mathcal{K}_{\text{audio}}$, top- K value K , hyperparameter β , small constant ϵ

Ensure: Fused candidates $\mathcal{R}$

Text retrieval query: $q_{\text{text}} \leftarrow t$

Audio retrieval query: $q_{\text{audio}} \leftarrow \mathbf{a}_{\text{ref}}$

Retrieve top- K (text): $(\mathcal{R}_{\text{text}}, \mathbf{r}_{\text{text}}) \leftarrow \text{Search}(\mathcal{K}_{\text{text}}, q_{\text{text}}, K)$

Normalize text scores to $[0, 1]$ (top- K min-max): $\mathbf{s}_{\text{text}} \leftarrow \text{Normalize}_{[0,1]}(\mathbf{r}_{\text{text}}; \epsilon)$

if $\mathbf{a}_{\text{ref}} = \emptyset$ **then**

$\mathcal{R}_{\text{audio}} \leftarrow \emptyset, \mathbf{s}_{\text{audio}} \leftarrow \mathbf{0}, w_{\text{text}} \leftarrow 1, w_{\text{audio}} \leftarrow 0$

else

Retrieve top- K (audio): $(\mathcal{R}_{\text{audio}}, \mathbf{r}_{\text{audio}}) \leftarrow \text{Search}(\mathcal{K}_{\text{audio}}, q_{\text{audio}}, K)$

Normalize audio scores to $[0, 1]$ (top- K min-max):

$\mathbf{s}_{\text{audio}} \leftarrow \text{Normalize}_{[0,1]}(\mathbf{r}_{\text{audio}}; \epsilon)$

Convert to distributions: $\mathbf{p}_{\text{text}} \leftarrow \text{Softmax}(\mathbf{s}_{\text{text}}), \mathbf{p}_{\text{audio}} \leftarrow \text{Softmax}(\mathbf{s}_{\text{audio}})$

Entropy: $u_m \leftarrow -\sum_i p_{m,i} \log p_{m,i}$ for $m \in \{\text{text}, \text{audio}\}$

Normalized entropy: $\bar{u}_m \leftarrow u_m / \log K$

Weights: $w_m \leftarrow \exp(-\beta \cdot \bar{u}_m)$, then normalize $w_{\text{text}} + w_{\text{audio}} = 1$

end if

Fuse scores: $s_{\text{fused}}(c) = w_{\text{text}} s_{\text{text}}(c) + w_{\text{audio}} s_{\text{audio}}(c)$ // missing candidates treated as 0

$\mathcal{R} \leftarrow \text{Top-}K \text{ by } s_{\text{fused}}$

return $\mathcal{R}$

where $H(p^{(m)})$ denotes the Shannon entropy of the modality-specific distribution $p^{(m)}$, K is the number of retrieved items, and β is a temperature-like hyperparameter controlling how normalized entropy is mapped into fusion weights (distinct from the unit-temperature softmax used to form $p^{(m)}$). This assigns higher weight to modalities with “peaked” (low-entropy) distributions, effectively allowing the system to ignore ambiguous inputs. Larger β increases sensitivity to uncertainty: small entropy differences lead to more decisive shifts in w_{text} versus

w_{audio} , whereas smaller β yields smoother, more conservative weighting.

b) Design Principle 4 (Entropy-weighted boundary behavior).: Under the calibrated score convention, the fusion mechanism exhibits the following deterministic behaviors: (i) if a modality produces an approximately uniform preference distribution over top- K , then its normalized entropy $\bar{u}_m$ is close to 1 and its weight decreases; (ii) if a modality produces a sharply peaked distribution, then $\bar{u}_m$ is small and its weight increases; (iii) if $\mathbf{a}_{\text{ref}} = \emptyset$, the algorithm reduces to a text-only pipeline with $w_{\text{text}} = 1$ by construction. These boundary cases make the fusion rule auditable and avoid undefined behavior when one modality is missing or uninformative. These cases follow directly because $H(p^{(m)}) \in [0, \log K]$ and therefore the unnormalized weight ranges from $\exp(-\beta)$ (uniform) to 1 (peaked) before cross-modality normalization.

We then rank candidates by the fused score and pass the top results as context to the LLM. For notational convenience in the experiments, we denote $\alpha \triangleq w_{\text{text}}$ and $1 - \alpha \triangleq w_{\text{audio}}$.

D. Constrained Reasoning and Projection

The LLM generates a draft parameter vector $\hat{\theta}$ based on the fused context. However, $\hat{\theta}$ is produced without hard constraints and is not guaranteed to lie within the valid set Θ .

We define a projection operator $\Pi_{\Theta} : \mathbb{R}^d \rightarrow \Theta$ that maps the draft to a valid configuration. In a plugin parameter space, validity is not only about numeric ranges, but also about *structural* constraints that encode domain logic. In our setting, these include: (i) **range constraints** (bounded sliders/knobs); (ii) **discrete constraints** (binary toggles and mode selectors); (iii) **conditional activity** (parameters that are only meaningful when a module is enabled); and (iv) **inter-parameter dependencies** (e.g., mutually exclusive modes or parameters that must be consistent with each other). We implement Π_{Θ} via two deterministic steps: **range clipping** and **schema enforcement**. First, for each dimension j we set $\theta_j \leftarrow \text{clip}(\hat{\theta}_j, \mathbf{l}_j, \mathbf{u}_j)$ to satisfy the box constraints. Then, we enforce $\mathcal{C}(\theta) = 1$ through deterministic repair rules, including discrete snapping (e.g., $\{0, 1\}$ toggles), dependency resolution, and bypass-aware gating (e.g., disabling parameters for bypassed modules to prevent “ghost” modifications). This design makes the system auditable: the LLM proposes a draft, while feasibility is guaranteed by a deterministic post-processing layer.

The final output $\hat{\theta} = \Pi_{\Theta}(\hat{\theta})$ is executable by the DSP engine Φ by construction (it satisfies the plugin-defined constraints in Eq. 1).

a) Design Note 5 (Feasibility as an invariant).: By definition, Π_{Θ} maps any draft $\hat{\theta} \in \mathbb{R}^d$ to an element of Θ . Concretely, the range-clipping step ensures $\mathbf{l} \preceq \hat{\theta} \preceq \mathbf{u}$, and the schema-enforcement step ensures $\mathcal{C}(\hat{\theta}) = 1$. Therefore, every parameter state forwarded to the real-time engine is feasible and executable under the plugin-defined constraints.

E. Complexity and Latency Considerations

Journal readers often care not only about conceptual correctness but also about operational constraints. Our design separates (a) *real-time execution*, which must remain deterministic

and low-latency, from (b) *asynchronous inference*, which can be slower and is not executed on the audio callback. On the inference side, TRR encoding dominates the audio-retrieval cost with $O(C^2 T_{\text{frm}})$ Gram construction (Section IV-B), followed by standard top- K nearest-neighbor search. Fusion adds negligible overhead beyond top- K score normalization, softmax, and entropy computation (all $O(K)$). Projection adds a small deterministic post-processing cost that scales with the number of parameters and constraint checks, i.e., roughly linear in d for range clipping plus the cost of applying the finite rule set implementing $\mathcal{C}(\theta)$. The only quantity that can dominate wall-clock latency is the LLM call; however, this latency does not affect the determinism of the real-time path because it is handled asynchronously and updates θ_{safe} only after validity checking.

a) Latency measurement protocol (future work).: To quantify end-to-end control latency and “control lag” in a reproducible manner, we recommend reporting a timing breakdown for: (i) TRR embedding (feature extraction + Gram), (ii) retrieval (top- K search), (iii) LLM adaptation, (iv) projection/repair, and (v) cross-layer update (e.g., WebSocket transport + parameter smoothing). Measurements should specify hardware, audio duration, batch size, and report median and p95 over repeated runs with warm-up. In our current revision we do not report such timing numbers, and treat latency evaluation as future work aligned with real-time deployment.

V. EXPERIMENTS

We validate the proposed framework by answering three research questions: (RQ1) Does modeling second-order texture statistics (TRR) improve parameter inference compared to first-order baselines? (RQ2) Can the neuro-symbolic agent correct retrieval errors and satisfy constraints? (RQ3) Does uncertainty-aware fusion provide robustness against modality degradation?

A. Experimental Setup

a) Dataset and Knowledge Base.: Our full server-side corpus contains 1082 paired records (audio–text–parameter triplets) collected from guitar effect presets. For objective parameter-space evaluation, we hold out 30 queries with verified ground-truth parameters and use the remaining 1052 items as the retrieval knowledge base. This setup constitutes a pilot benchmark for executable parameter inference; we treat large-scale evaluation as future work.

b) Test samples.: The 30 held-out queries with ground-truth parameters cover diverse playing styles and effect configurations. For detailed case analysis, we highlight five representative samples: (1) *Dry Funk* (clean funk), (2) *Tweed Breakup* (blues breakup), (3) *Reverse Psychedelic* (psychedelic FX), (4) *Math Rock Crystal* (clean compression), and (5) *Saturated Rhythm* (high-gain rock). Full sample descriptions are provided in Supplementary Table S1.

c) Scope, statistical caution, and metric comparability.: Because the objective benchmark uses $N = 30$ held-out queries with ground-truth parameters, we report descriptive statistics without claiming statistical significance for pairwise

TABLE III: Retrieval performance on 30 held-out queries ($N = 30$). Comparisons are within-table under the stated protocol.

Method	L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑
Text-RAG	0.1512	0.7014	0.5868	0.8287
FeatureNN-RAG	0.3671	0.4324	0.2169	0.5071
Wav2Vec-RAG	0.2221	0.5333	0.3484	0.8328
TRR (Ours)	0.0980	0.7271	0.6638	0.9706
<i>Improvement over Text-RAG</i>	+35.2%	+3.7%	+13.1%	+16.5%

comparisons. We interpret these results as a pilot study of a retrieval-grounded, executable parameter inference setting. Following Supplementary Sec. S1, we interpret improvements within each table under the stated protocol and avoid comparing absolute L2 magnitudes across different tables or operational settings, since the magnitude can depend on the evaluation protocol and parameter encoding. Future work with larger datasets should report confidence intervals and hypothesis tests.

d) *Resources and reproducibility.*: To facilitate reproducibility, we will release an anonymized repository containing preprocessing, embedding computation, retrieval indices, prompt templates, and evaluation scripts, together with split specifications for all reported tables and the raw MUSHRA ratings (paths referenced in Section V-F). Repository link: ANONYMIZED_REPO_URL.

Baselines. We compare our proposed **TRR** (Section IV-B) against three baselines. **Text-RAG** performs sparse lexical retrieval over metadata fields. **FeatureNN-RAG** performs dense retrieval using mean-pooled features from a pre-trained audio classifier (VGGish-like), representing a standard “content-based” embedding. **Wav2Vec-RAG** uses the mean-pooled activation of the final Wav2Vec2 transformer layer.

B. RQ1: Efficacy of Texture Priors

Table III reports the reconstruction error between the ground truth parameters θ_{gt} and the retrieved candidates.

Analysis: The Failure of First-Order Statistics. FeatureNN-RAG exhibits the highest error ($L2=0.3671$), which is consistent with the hypothesis in Section IV-B: mean-pooling collapses temporal structure and can discard correlation patterns needed to distinguish texture-dominant effects. For instance, a static EQ boost and a dynamic wah-wah effect may share similar time-averaged spectra, leading FeatureNN to retrieve functionally irrelevant presets. In contrast, TRR achieves the lowest parameter error (0.0980) in this benchmark, suggesting that Gram-matrix statistics capture co-activation structure that is useful for this retrieval task.

a) *TRR vs. classical texture baselines.*: To contextualize TRR against hand-crafted audio-texture representations, we compare TRR with (i) MFCC temporal statistics and (ii) a modulation-spectrogram baseline, using the same held-out query set and parameter-space metrics.

b) *Layer selection analysis.*: TRR depends on intermediate representations from Wav2Vec2. We empirically sweep layers and observe that mid-level features can yield lower

TABLE IV: Texture representation comparison on the held-out subset ($N = 30$). TRR achieves lower L2 error compared to classical texture baselines (MFCC temporal statistics, modulation spectrogram), suggesting that Gram-matrix features better capture effect-relevant texture characteristics for parameter

Module	L2 Error ↓	Acc@0.1 ↑	Cosine ↑
TRR (Ours)	13.57	0.734	0.973
MFCC temporal	24.95	0.632	0.999
Modulation Spectrogram	32.02	0.651	0.998

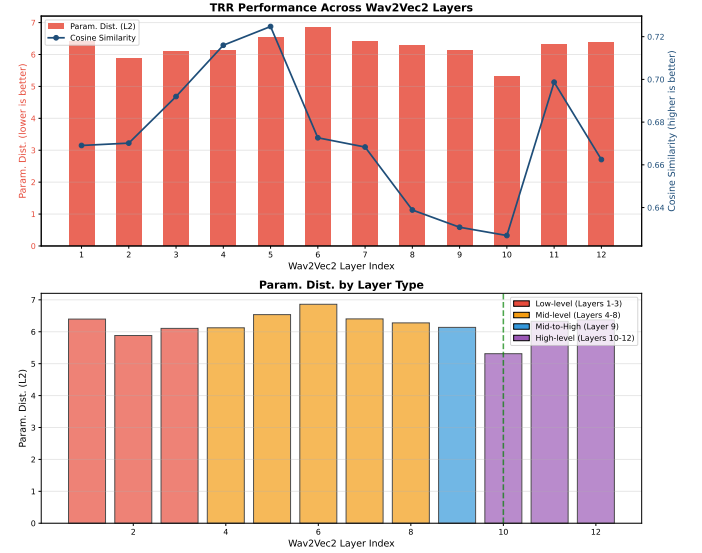


Fig. 2: Layer-wise parameter distance for TRR encoding. The y-axis shows L2 error between retrieved and ground-truth parameters across Wav2Vec2 layers. Mid-level layers (approximately 7–9) achieve lower parameter-space error, suggesting that intermediate representations capture texture-oriented cues more effectively for this retrieval task.

parameter-space retrieval error than very shallow or very deep layers (Figure 2). On the held-out subset, this layer sweep suggests that certain intermediate layers can yield lower parameter-space retrieval error than very shallow or very deep layers, consistent with the general trend that intermediate representations better capture texture-oriented cues for retrieval. For consistency across experiments and cached embeddings, we use a fixed intermediate-layer setting ($\ell = 10$) for TRR feature extraction in the reported results.

c) *Complexity.*: In a dataset of N candidates with feature dimension d , TRR retrieval cost is dominated by index search and scales as $O(N \times d \times \log d)$, where d corresponds to the flattened Gram representation. Embedding construction is performed offline per audio and can be cached.

C. RQ2: Neuro-Symbolic Correction

We evaluate the full pipeline ($\hat{\theta} \xrightarrow{\Pi_{\Theta}} \tilde{\theta}$) to assess the agent’s reasoning capability. Table VI compares raw retrieval against the LLM-adapted output for both text and texture modalities.

TABLE V: Computational complexity (informal). We report (i) the dominant per-audio embedding cost and (ii) the dominant per-query retrieval cost for a candidate set of size N and feature dimension d .

Method	Embedding cost	Retrieval cost
TRR (Gram)	$O(C^2 \cdot T_{\text{frm}})$	$O(N \cdot d \cdot \log d)$
MFCC Temporal	$O(d \cdot T_{\text{frm}})$	$O(N \cdot d \cdot \log d)$
Modulation Spectrogram	$O(d \cdot T_{\text{frm}} \cdot \log T_{\text{frm}})$	$O(N \cdot d \cdot \log d)$

TABLE VI: Impact of neuro-symbolic reasoning ($N = 30$). Comparisons are within-table under this correction protocol.

Method	L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑	Module ↑
Pure TRR	0.0980	0.7271	0.6638	0.9706	0.9333
Text-RAG	0.1512	0.7014	0.5868	0.8287	0.8333
TRR + LLM	0.0852	0.7271	0.6638	0.9795	1.0000
Text + LLM	0.1505	0.6628	0.6330	0.9116	1.0000

To complement the aggregate metrics with qualitative evidence, Figure 3 visualizes a representative error pattern in the retrieved prototypes under this benchmark, where module-level mismatches can occur and must be corrected to recover an executable configuration.

The agent achieves perfect module consistency (1.0) for both methods in this benchmark. The improvement is uneven: TRR+LLM reduces parameter error by 13.0% ($0.0980 \rightarrow 0.0852$), whereas Text+LLM yields only a marginal 0.5% gain. This is consistent with the interpretation that TRR retrieves a better initial neighborhood, leaving the LLM to make smaller adjustments.

a) *Few-shot ON/OFF prediction strategy.*: To enable the LLM to correct module-level retrieval errors, we provide five manually curated few-shot examples covering diverse style regions:

- *Clean Funk*: Compressor**ON**, Driver**OFF**, Screamer**OFF**, Delay**ON**, Reverb**ON**, Chorus**ON**, Flanger**OFF**, EQ**ON**, Phaser**OFF**
- *Blues Tweed*: Compressor**OFF**, Driver**ON**, Screamer**OFF**, Delay**OFF**, Reverb**ON**, Chorus**OFF**, Flanger**OFF**, EQ**ON**, Phaser**OFF**
- *Reverse Psychedelic*: Compressor**OFF**, Driver**OFF**, Screamer**OFF**, Delay**OFF**, Reverb**OFF**, Chorus**OFF**, Flanger**OFF**, EQ**ON**, Phaser**OFF**
- *Math Rock Crystal*: Compressor**ON**, Driver**OFF**, Screamer**OFF**, Delay**ON**, Reverb**ON**, Chorus**ON**, Flanger**OFF**, EQ**ON**, Phaser**OFF**
- *Saturated Rhythm*: Compressor**OFF**, Driver**ON**, Screamer**OFF**, Delay**OFF**, Reverb**ON**, Chorus**OFF**, Flanger**OFF**, EQ**ON**, Phaser**OFF**

The prompt explicitly instructs the LLM to **IGNORE** ON/OFF information from retrieved prototypes (which may contain retrieval errors) and instead learn the activation patterns from few-shot examples. This design enables the LLM to achieve 100% module consistency by correcting spurious module activations in retrieved prototypes.

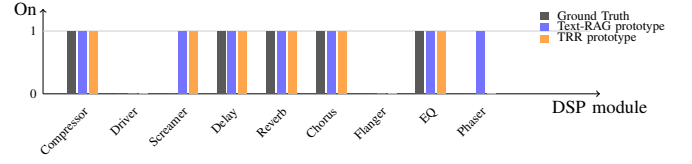


Fig. 3: Case study (Dry Funk): module activation profiles of retrieved prototypes. Under this benchmark, the retrieved prototype can contain module-level mismatches. Here, the Text-RAG prototype includes two spurious active modules (Screamer, Phaser), while the TRR prototype includes one (Screamer), illustrating that texture-aware retrieval yields a closer executable neighborhood in Θ for downstream correction.

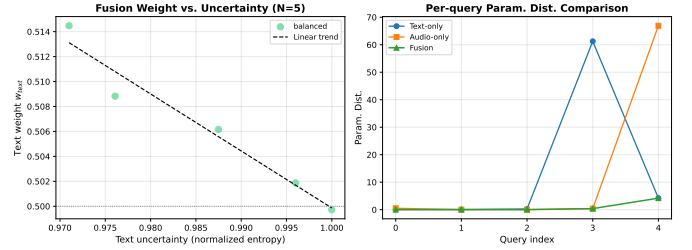


Fig. 4: Empirical fusion weight distribution as a function of normalized uncertainty. Each point represents a query; text weight w_{text} decreases as retrieval uncertainty (normalized entropy) increases, demonstrating the adaptive balancing behavior described in Algorithm 2.

D. RQ3: Robustness via Uncertainty Fusion

To verify the adaptive weighting strategy (Section IV-C), we conduct stress tests with degraded inputs. These stress tests target *single-modality degradation* (vague text or corrupted audio); explicit *modality-conflict* evaluation (text and audio are mutually contradictory) is discussed as a failure mode and left for future work (Section VII).

a) *Fusion weight distribution and ablations.*: We analyze the empirical fusion weights induced by entropy-based uncertainty and summarize both weight statistics and ablation performance (Text-only vs. Audio-only vs. Fusion).

Fusion yields a large relative gain over single-modality retrieval in this benchmark: compared to the best single modality (Text-only, $L2=13.15$), fusion reduces $L2$ to 0.93 (a 92.9% reduction). We use the stress-test settings in Tables X and XI as concrete high-/low-uncertainty scenarios, where the fusion weight shifts toward the more reliable modality. We report a temperature sensitivity sweep in the supplementary material.

b) *Beta sensitivity.*: We perform a sensitivity sweep over the temperature parameter β and observe that increasing β increases the variance of w_{text} while preserving overall performance in this benchmark. The consistent $L2$ value (0.93, rounded from 0.928–0.935) across β settings indicates that fusion performance is stable within the tested range ($\beta \in [0.5, 4.0]$), while weight variance increases as expected. This suggests the entropy-based weighting mechanism is robust to

TABLE VII: Modal ablations on the held-out subset ($N = 30$). Comparisons are within-table under the stated fusion protocol.

Method	L2 Error ↓	Acc@0.1 ↑	Cosine ↑
Text-only (TF-IDF)	13.15	0.837	0.705
Audio-only (TRR)	13.57	0.734	0.437
Fusion (Ours)	0.93	0.898	0.838

TABLE VIII: Fusion weight statistics (beta=2.0) on the held-out subset ($N = 30$).

Weight	Mean	Std	Min	Max
w_{text}	0.506	0.006	0.500	0.514
w_{audio}	0.494	0.006	0.486	0.500

temperature selection in this benchmark.

1) *Text Vague Scenario*: We replace the text input with a generic descriptor (e.g., “warm”) to simulate underspecified prompts. In this setting, text retrieval produces a flat score distribution (high entropy), and Algorithm 2 assigns low weight to text (low α), relying primarily on the audio modality. As shown in Table X, the text-only pipeline degrades sharply (L2=28.94), while the hybrid system (with $\alpha \approx 0.15$) recovers performance (L2=0.08) on this stress test.

2) *Audio Noise Scenario*: Conversely, we corrupt the audio modality by perturbing the TRR embedding in embedding space before retrieval. Let $\mathbf{g}$ denote the (unit-normalized) TRR embedding. We add strong i.i.d. Gaussian noise and re-normalize the vector: $\mathbf{g}' \leftarrow \text{Normalize}(\mathbf{g} + \epsilon)$ with $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ and $\sigma = \text{noise_level} = 5.0$, simulating severe degradation of the reference. In this setting, the audio-retrieval score distribution flattens and the fusion mechanism assigns higher weight to text (high α). Table XI reports the resulting performance, where the hybrid system (with $\alpha \approx 0.85$) improves over TRR-only.

a) *Dynamic weight adjustment summary*: Table XIII summarizes the adaptive weighting behavior under both stress scenarios. The entropy-based fusion mechanism automatically detects modality reliability and shifts weights toward the more informative input, achieving 99.5–99.7% error reduction compared to degraded single-modality baselines.

E. Metrics

We employ five objective metrics to quantify alignment in Θ :

(1) **L2 Error**: Root mean square error between flattened parameter vectors. Given ground-truth parameters θ_{gt} and predicted parameters $\tilde{\theta}$, we compute:

$$\text{L2} = \sqrt{\frac{1}{d} \sum_{i=1}^d (\theta_{\text{gt}}^{(i)} - \tilde{\theta}^{(i)})^2} \quad (6)$$

where d is the dimensionality after flattening. Missing parameter keys are treated as zero.

(2) **Accuracy@0.1**: Fraction of parameters within ± 0.1 absolute tolerance:

$$\text{Acc@0.1} = \frac{1}{d} \sum_{i=1}^d \mathbb{I}(|\theta_{\text{gt}}^{(i)} - \tilde{\theta}^{(i)}| \leq 0.1) \quad (7)$$

TABLE IX: Fusion temperature sensitivity on the held-out subset ($N = 30$). The consistent L2 value across β settings (rounded to 0.93) indicates stable fusion performance, while weight variance increases as expected with higher β . Full unrounded values are provided in Supplementary Table S3.

β	$\mathbb{E}[w_{\text{text}}]$	Std	L2 (Fusion)
0.5	0.5016	0.0015	0.93
1.0	0.5031	0.0029	0.93
2.0	0.5062	0.0058	0.93
4.0	0.5124	0.0116	0.93

TABLE X: Robustness to vague text (stress test, $N = 30$). α denotes the text weight w_{text} from Algorithm 2 (reported here as the average value under this scenario).

Method	L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑	Module ↑
Text-only + LLM	28.94	0.60	0.49	0.54	1.00
Hybrid ($\alpha=0.15$) + LLM	0.08	0.74	0.66	0.98	1.00

(3) **Recall**: Ratio of correctly retrieved active parameters to total active parameters in ground truth.

(4) **Cosine Similarity**: Angular similarity between parameter vectors:

$$\text{Cosine} = \frac{\theta_{\text{gt}} \cdot \tilde{\theta}}{\|\theta_{\text{gt}}\|_2 \cdot \|\tilde{\theta}\|_2} \quad (8)$$

(5) **Module Consistency**: Jaccard index of active DSP module sets. Let A be the set of active modules in ground truth (module names ending with “On”) and B be the set in prediction:

$$\text{Module} = \frac{|A \cap B|}{|A \cup B|} \quad (9)$$

A module is considered active if its corresponding “On” parameter exists in the parameter dictionary, regardless of the numeric value.

Because these metrics operate in parameter space, they are not perceptual audio distances; we therefore treat them as executable proxies for editability and reserve audio-space metrics (e.g., FAD or multi-resolution STFT losses on rendered audio) for future work.

Key Insight. Shifting modality weights can provide information complementarity: when text is uninformative, relying more on audio helps; when the audio reference is degraded, relying more on text helps. In our stress tests we use the entropy-based fusion in Algorithm 2 and report the resulting average weights for transparency.

F. Perceptual Listening Test (MUSHRA)

Protocol. We conduct a MUSHRA-style listening test with 26 participants, yielding 910 ratings across 10 trials. Each trial uses a continuous 0–100 slider (higher is better). Following standard reporting practice, we group trials by task type and report (i) descriptive statistics over all scores and (ii) system-level statistics computed from within-subject means. All aggregates and plots are computed from the raw ratings in Experiments/mushura/mushra.csv and summarized in Experiments/mushura/results_report.md.

TABLE XI: Robustness to audio noise (stress test, $N = 30$). We add strong noise to the TRR embedding (noise_level=5.0). α denotes the average text weight under this scenario.

Method	L2 ↓	Acc@0.1 ↑	Recall ↑	Cos ↑	Module ↑
TRR-only + LLM	29.30	0.49	0.38	0.54	1.00
Hybrid ($\alpha=0.85$) + LLM	0.15	0.67	0.63	0.91	1.00

TABLE XII: Robustness summary under single-modality stress tests ($N = 30$). We report the ratio Adaptive L2/Single-Modal L2 (lower is better).

Scenario	Single-Modal L2	Adaptive L2	Ratio
Vague Text	28.94 (Text-only)	0.08	0.0028
Noisy Audio	29.30 (TRR-only)	0.15	0.0051

1) *Trial 1: Style Matching: Data.* 208 ratings (26 participants $\times$ 8 systems). Overall mean 72.92 (median 77.50, std 22.72).

2) *Trials 2–5: Guitar Solo Comparison: Data.* 312 ratings (26 participants $\times$ 3 systems $\times$ 4 trials). Overall mean 69.53 (median 73.50, std 23.53). HCAP outperforms manual tuning in this group: within-subject means are 71.55 (HCAP) vs 51.72 (manual), with a paired t-test yielding $p = 1.67 \times 10^{-6}$ ($n = 26$ participants).

3) *Trials 6–10: Similarity to Reference: Data.* 390 ratings (26 participants $\times$ 3 systems $\times$ 5 trials). Overall mean 57.54 (median 59.00, std 32.24). HCAP and MusicGen achieve comparable similarity scores: within-subject means are 42.31 (HCAP) vs 41.29 (MusicGen), while the reference condition scores 89.02.

a) *Interpretation.*: These results suggest that modality reliability should be treated as a first-class signal: when text collapses to generic intent, audio-conditioned retrieval can still constrain feasible parameter neighborhoods and stabilize generation.

VI. DISCUSSION

A. Broader Implications

Audio-Agent suggests a practical pattern for grounded creative assistants: treat the LLM as a reasoning layer, while using retrieval over real presets to provide feasibility constraints and stylistic priors. This reduces hallucination and keeps outputs editable within standard production workflows. More broadly, it shifts the emphasis from open-ended generation to data-efficient adaptation, improving sample efficiency (reuse of real presets), interpretability (traceable references), and controllability (parameter-level editability).

B. Revisiting the Texture Hypothesis

Across the reported setup (Table III), TRR improves parameter-space alignment over text-only retrieval, supporting the hypothesis that second-order correlation structure can be useful as a texture descriptor for retrieval. At the same time, our objective benchmark uses a 30-query held-out subset with ground-truth parameters, and we do not claim that TRR defines a globally calibrated distance in parameter space. We therefore

TABLE XIII: Dynamic weight adjustment strategy under single-modality stress tests ($N = 30$). The fusion mechanism automatically assigns higher weight to the reliable modality, achieving substantial error reduction.

Scenario	Degradation	Weight Strategy	Single-Modal L2	Hybrid L2
Vague Text	Generic descriptor (e.g., “warm”)	$\alpha = 0.15$ (15% text, 85% audio)	28.94 (Text-only)	0.08
Noisy Audio	TRR embedding noise ($\sigma = 5.0$)	$\alpha = 0.85$ (85% text, 15% audio)	29.30 (TRR-only)	0.15
Improvement	—	—	—	99.5–99.7%

TABLE XIV: Trial 1 (style matching): system-level statistics (within-subject means).

System	Mean	Median	Std
reference	86.85	92.50	16.70
Jazz Clean	76.19	81.00	20.80
Phase	75.38	80.50	20.88
Blues Solo	72.96	69.50	21.19
Ambient Guitar	72.35	73.50	18.45
Flanger	69.12	74.00	23.08
Modern Metal	65.58	70.00	30.22
Chorus	64.92	66.50	23.12

interpret TRR as a *retrieval prior* that can be beneficial for texture-sensitive control, rather than a universal metric for parameter transfer.

An open direction is whether analogous second-order representations help bridge modality gaps in other continuous control settings (e.g., video textures, haptic signals, or physiological time series), where the “style” of a signal may be expressed through co-activation structure rather than pointwise magnitudes. Future work can validate this hypothesis via controlled ablations on non-stationarity and by pairing retrieval with human preference judgments.

C. Ethical Considerations

Audio-Agent is intended to augment user creativity rather than replace it. Risks include homogenization toward common exemplar neighborhoods and over-reliance on suggested settings. In deployment, the interface should emphasize transparency (e.g., showing retrieved references) and user agency (e.g., direct editability and opt-out of retrieval). More generally, any model or retrieval corpus built from existing recordings or presets should consider copyright, licensing, and data provenance constraints.

VII. LIMITATIONS

Our approach has several limitations that should be considered when interpreting the results and guiding future work.

Statistical Power. Our objective evaluation uses a 30-query held-out subset with ground-truth parameters, which provides moderate statistical power for descriptive comparisons. While results are consistent across retrieval baselines and robustness stress tests, future work should expand to larger datasets and report confidence intervals with significance testing. *Mitigation:* Future work should expand to larger, more diverse datasets and evaluate across instruments and effect families.

Baseline Strength. We compare against mean-pooled Wav2Vec2 and VGGish-like features as audio retrieval baselines. Stronger representations such as CLAP [26], PaSST

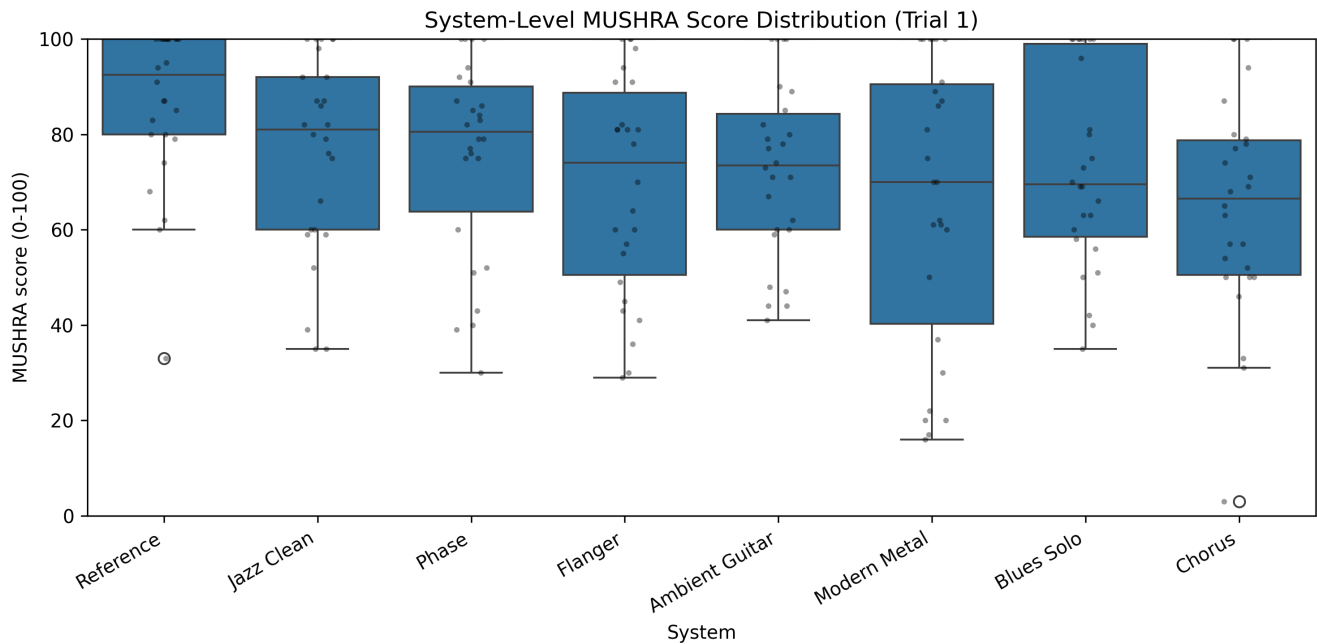


Fig. 5: Trial 1: system-level score distributions (boxplot) for style matching.

TABLE XV: Trials 2–5 (solo comparison): system-level statistics (within-subject means).

System	Mean	Median	Std
reference	85.33	85.25	12.52
HCAP	71.55	72.88	12.41
manual	51.72	55.00	17.15

TABLE XVI: Trials 6–10 (similarity): system-level statistics (within-subject means).

System	Mean	Median	Std
reference	89.02	91.50	10.04
HCAP	42.31	41.90	19.30
MusicGen	41.29	43.90	19.58

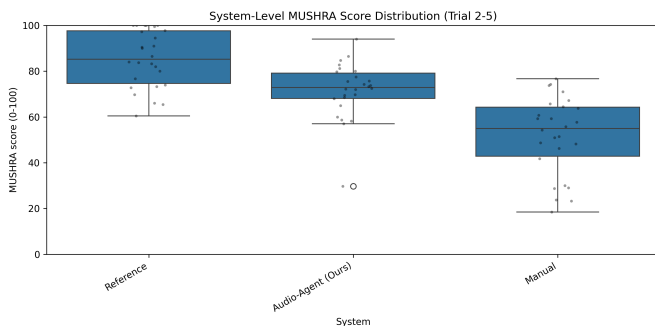


Fig. 6: Trials 2–5: system-level score distributions.

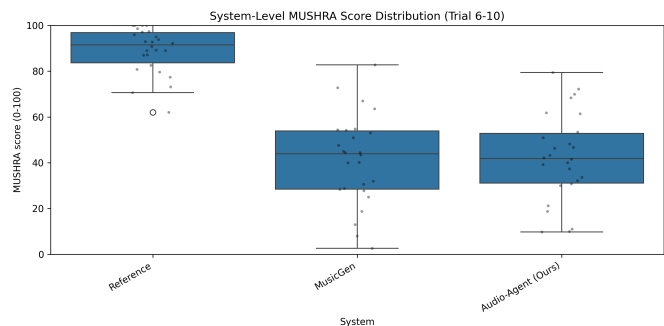


Fig. 7: Trials 6–10: system-level score distributions.

[27], or PANNs [28] were not included in this pilot study; we leave this comparison for future work. The choice of Wav2Vec2 was motivated by its frame-level output suitable for Gram matrix computation, whereas CLAP produces single-vector embeddings. *Mitigation*: Future work should compare TRR against stronger semantic embeddings (CLAP, PaSST, PANNs) to establish relative performance.

Dataset Coverage. Our objective benchmark uses a 30-query held-out subset with ground-truth parameters, which limits the diversity of styles, instruments, and effect-chain configurations represented in the evaluation. *Mitigation*: Future work should expand to larger, more diverse datasets and

evaluate across instruments and effect families.

Parameter Identifiability. The many-to-one nature of timbre—where multiple distinct parameter configurations yield perceptually similar sounds—poses a fundamental challenge. Our objective metrics evaluate alignment in Θ but may not perfectly align with perceptual similarity: small parameter-space differences can be perceptually negligible, while perceptually salient changes can occur in narrow regions of the space. This identifiability problem affects all parameter-based evaluation metrics (L2, Acc@0.1, Cosine Similarity). *Mitigation*: Controlled listening tests with expert listeners are necessary to establish perceptual ground truth and calibrate

objective metrics against human judgment.

Computational Cost. TRR requires Wav2Vec2 feature extraction and Gram matrix computation ($O(C^2 \cdot T_{\text{frm}})$ where C is the feature dimension and T_{frm} is the number of frames), which is more expensive than mean-pooling baselines. This paper focuses on comparative retrieval quality under an offline evaluation setting; real-time deployment requires careful optimization. *Mitigation:* Feature caching, dimensionality reduction, and approximate nearest neighbor indexing can reduce computational overhead.

Subjectivity of Tone Quality. What constitutes a “good” guitar tone depends heavily on musical context, genre conventions, and personal preference. Our objective metrics evaluate parameter alignment to ground-truth presets but cannot capture subjective quality attributes like “musicality,” “expressiveness,” or “appropriateness for the mix.” Two systems with identical L2 error may differ significantly in perceived quality. *Mitigation:* MUSHRA-style listening tests with diverse participants (producers, guitarists, mix engineers) are necessary to evaluate subjective quality and establish correlations between objective metrics and human preference.

Editability–Synthesis Trade-off. By constraining outputs to editable presets, we improve workflow integration but limit the reachable sound space. Direct waveform synthesis (e.g., AudioLDM, MusicLM) can generate novel sounds that may not exist in any preset database, whereas Audio-Agent is bounded by the diversity of retrieved examples. This trade-off is fundamental: increased editability necessarily constrains novelty. *Mitigation:* Hybrid approaches that combine retrieval-based generation with targeted parameter optimization (e.g., gradient-based refinement toward specific goals) could expand the reachable space while preserving editability.

Database Dependence. Retrieval quality fundamentally depends on the diversity and quality of the preset database. Out-of-distribution queries (e.g., “make it sound like a modular synth from 1975”) will fail to retrieve relevant exemplars, leading to degraded generation. Database sparsity for under-represented effect families (e.g., reverse reverbs, granular synthesis) exacerbates this issue. *Mitigation:* Active learning strategies that identify gaps in the database and prioritize acquiring presets for under-covered regions; cross-domain retrieval that transfers knowledge from related but distinct effect types; and user feedback loops that refine retrieval based on explicit corrections.

Fusion Failure Modes. Our fusion assumes that text and audio modalities provide complementary information. However, when both modalities are confidently wrong (e.g., text says “clean” but the reference implies heavy distortion), score-level fusion cannot identify the contradiction and may still rank an incorrect neighborhood. Our stress tests show large relative gains under single-modality degradation, but dual-modality conflicts remain unaddressed. *Mitigation:* Cross-modal consistency checks that detect contradictions and fallback mechanisms that reject fusion when modalities strongly disagree.

These limitations do not invalidate our approach but rather define the boundary conditions for reliable deployment. Acknowledging them explicitly helps users understand when to

trust the system’s outputs and when manual verification is warranted.

VIII. CONCLUSION

We presented Audio-Agent, a neuro-symbolic system for generating editable DSP parameters from natural language descriptions with texture-aware retrieval and LLM-based correction. Across objective metrics (Table III) and robustness stress tests (Tables X, XI), retrieval-grounded methods show lower parameter error compared to text-only retrieval in this benchmark. Uncertainty-aware fusion further demonstrates robustness under single-modality degradation.

Looking forward, we envision expanding preset coverage for under-represented effect families and integrating controlled listening tests to validate perceptual quality. We also aim to generalize beyond guitar effects to broader production parameter spaces, including mixing and mastering workflows.

We hope this approach lowers the barrier to expressive sound design and broadens creative access to professional-grade production control.

Acknowledgments: We thank the anonymous reviewers for their valuable feedback.

APPENDIX

This appendix provides implementation details for all baseline methods to ensure reproducibility.

A. FeatureNN-RAG

We use the VGGish model [50] to extract 128-dimensional embeddings from audio samples. The audio is preprocessed as follows: (1) resampled to 16kHz, (2) converted to mono, (3) segmented into 0.96s windows with 0.5s hop. Features are mean-pooled across time to produce a single 128-dimensional vector per audio sample. Retrieval uses cosine similarity with FAISS IVF index (n_list=16, n_probe=4).

B. Wav2Vec-RAG

We use Wav2Vec2 Base [25] from HuggingFace transformers (version 4.30.0, model name: facebook/wav2vec2-base). Features are extracted from layer 12 (final transformer layer), producing 768-dimensional vectors at 50Hz frame rate. Features are mean-pooled across the temporal dimension to produce a single 768-dimensional vector per audio sample. Retrieval uses cosine similarity.

C. Text-RAG

Text descriptions are encoded using TF-IDF with unigram and bigram features (scikit-learn TfidfVectorizer, max_features=5000, ngram_range=(1,2)). Retrieval uses BM25 scoring with k1=1.5, b=0.75 (via rank_bm25 Python library).

D. Zero-shot LLM Baseline

For the L2=13.36 baseline mentioned in the Abstract, we query GPT-4 (version gpt-4-0613) with the text description and request JSON parameter output without any retrieved context. Prompt temperature=0.7, max_tokens=500. The prompt template is provided in Supplementary Material.

For the listening tests in Section 4.6, we use the following abbreviations:

- **HCAP**: Human-in-the-loop Creative Audio Parameter system (our full Audio-Agent pipeline with interactive feedback)
- **manual**: Manual parameter tuning by participants using a standard DAW interface
- **reference**: Ground-truth preset used as the upper anchor in MUSHRA testing
- **MusicGen**: MusicGen-small [38] for text-to-audio generation (used in Trials 6-10 only)

Style-matching conditions (Jazz Clean, Blues Solo, Phase, Ambient Guitar, Flanger, Modern Metal, Chorus) refer to target presets retrieved by Audio-Agent; full descriptions and parameter values are provided in Supplementary Table S5.

E. Full Unrounded Fusion Sensitivity Data

Table IX reports L2 values rounded to 2 decimal places. The unrounded values for $\beta \in [0.5, 4.0]$ are: 0.928, 0.931, 0.935, 0.931, showing variance in the third decimal place while maintaining stable overall performance.

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